

Brazilian E-Commerce Public Dataset by Olist

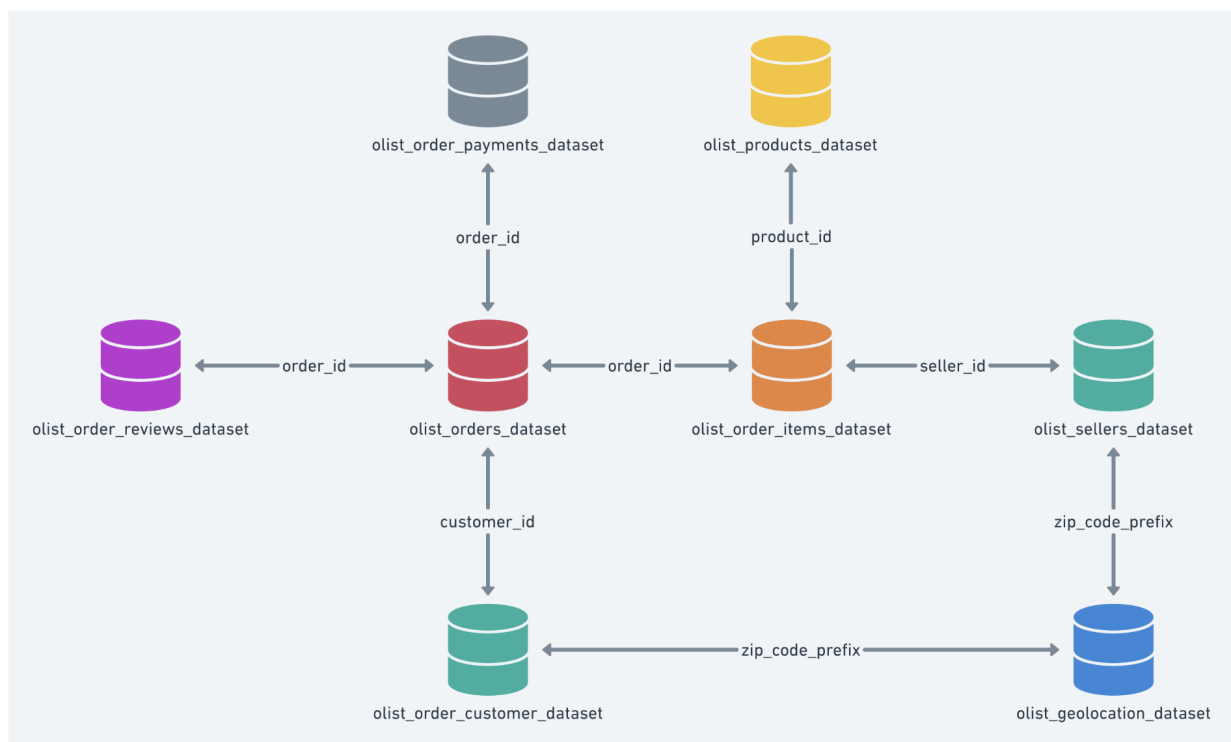
This is the dataset we will use to practice the complete journey from raw local files to a structured database, data analysis, cleaning, feature engineering, and supervised machine learning.

Relational Data

Multiple CSV Files

EDA and SQL

Classification /
Regression



Simplified structure of the main tables and their relationships.

Dataset page

Open the Kaggle page to download the files, read the full description, inspect the columns, and browse community notebooks:

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

WHAT IS INSIDE?

A real e-commerce data model

The data is not provided as one ready-made machine learning table. It is split into related files representing orders, customers, products, sellers, payments, reviews, and locations.

File	What one row represents
<code>olist_orders_dataset.csv</code>	One order and its purchase, approval, delivery, and estimated delivery timestamps.
<code>olist_customers_dataset.csv</code>	One customer record linked to an order, including city, state, and ZIP-code prefix.
<code>olist_order_items_dataset.csv</code>	One item inside an order, including product, seller, price, and freight value.
<code>olist_order_payments_dataset.csv</code>	One payment record for an order, including payment type and installments.
<code>olist_order_reviews_dataset.csv</code>	One customer review, score, and optional review text.
<code>olist_products_dataset.csv</code>	One product, its category, dimensions, weight, and description information.
<code>olist_sellers_dataset.csv</code>	One seller and its location.
<code>olist_geolocation_dataset.csv</code>	Geographical coordinates linked to Brazilian ZIP-code prefixes.
<code>product_category_name_translation.csv</code>	Translation of product categories from Portuguese to English.

order_id Orders, items, payments, reviews	customer_id Orders and customers	product_id Items and products	seller_id Items and sellers
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Important database lesson

The final ML dataset will usually need one row per order. Students must first aggregate tables such as order items and payments before joining them, otherwise values can be duplicated.

Important ML lesson

Students must decide when the prediction is made and avoid using future information. Reviews and actual delivery dates can be used for analysis, but may create leakage if used as model inputs.

WHAT CAN WE BUILD?

One dataset, several possible projects

Download locally

Load into a local database

Explore and clean

Engineer an ML table

Train and evaluate a model

Delivery Performance

Predict whether an order will arrive after its estimated delivery date.

Customer Satisfaction

Classify whether an order is likely to receive a low review score.

Sales Prediction

Use purchase dates, products, and locations to predict future sales.

Product Quality

Study which categories and order characteristics are linked to dissatisfaction.

NLP

Analyze review text and connect it to products, delivery, and customer ratings.

Clustering and Feature Engineering

Create customer, seller, product, location, and order-level behavioral features.

Recommended first supervised problem

Late-delivery classification: predict whether a delivered order will arrive after the estimated delivery date. It is easy to understand, has a clear binary target, and requires students to use several related tables.

Before the practical work

- Open the Kaggle page and read the dataset description.
- Download and inspect the file names and columns.
- Understand the relationships shown in the schema.
- Browse a few public notebooks for ideas, not for copying.

Why this dataset fits the training

- It looks like real company data, not one clean spreadsheet.
- It requires database design, SQL joins, and aggregation.
- It contains meaningful cleaning and feature engineering work.
- It supports both classification and regression directions.

The exact modelling target and project requirements will be confirmed during the training.